Rapport-CSC8607-TP1-YU-Zhuoyan

Ex1

1.c

```
$ nvidia-smi
Thu Sep 17 08:20:49 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 595.84                 Driver Version: 595.84         CUDA Version: 13.2     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA L4                      Off |   00000000:06:10.0 Off |                    0 |
| N/A   34C    P8             16W /   72W |       3MiB /  23034MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

Nvidia L4

1.d

```
[controller] zyu1@arcadia-slurm-controller:~
$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
              1580       gpu     bash     zyu1  R       2:11      1 starfighter-slurm-node-01-1
[controller] zyu1@arcadia-slurm-controller:~
```

```
[controller] zyu1@arcadia-slurm-controller:~
$ scancel 1580
```

scancel 1580

1.e

```
x1 [controller] zyu1@arcadia-slurm-controller:~/TP1
$ sbatch hello.sh
Submitted batch job 1583
```

```
x2 [controller] zyu1@arcadia-slurm-controller:~/TP1
$ ls -la logs/
total 12
drwxr-xr-x+  2 zyu1 1769 4096 Sep 17 13:04 .
drwxr-xr-x+  3 zyu1 1769 4096 Sep 17 13:04 ..
-rw-r--r--+  1 zyu1 1769    0 Sep 17 13:04 hello-slurm-1583.err
-rw-r--r--+  1 zyu1 1769 1752 Sep 17 13:04 hello-slurm-1583.out
```

hello-slurm-1583.out

1.f

```
[controller] zyu1@arcadia-slurm-controller:~/TP1
$ sacct -j 1583 --format=JobID,State,Elapsed,MaxRSS,ReqMem,ReqCPUS
JobID           State    Elapsed     MaxRSS     ReqMem  ReqCPUS
------------ ---------- ---------- ---------- ---------- --------
1583          COMPLETED   00:00:01                    8G        1
1583.batch    COMPLETED   00:00:01     17960K                   1
1583.extern   COMPLETED   00:00:02                              1
```

ReqMem: requested memory

MaxRSS: maximum resident set sizemax, the max of memory that was used.

1.g

```
➜  Desktop scp tmp.md tsp-client:~/TP1/
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
tmp.md                                        100%    6     1.1KB/s   00:00
➜  Desktop scp tsp-client:~/TP1/hello.sh hello.sh
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
hello.sh                                      100%  347    31.1KB/s   00:00
➜  Desktop rsync -avhP TP1/ tsp-client:~/TP1/
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
rsync(63875): error: TP1/: (l)stat: No such file or directory
Transfer starting: 0 files

sent 29 bytes  received 20 bytes  466 bytes/sec
total size is 0  speedup is 0.00
rsync(63875): warning: child 63876 exited with status 23
➜  Desktop rsync -avhP TP1/ tsp-client:~/TP1/
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Transfer starting: 1 files
./

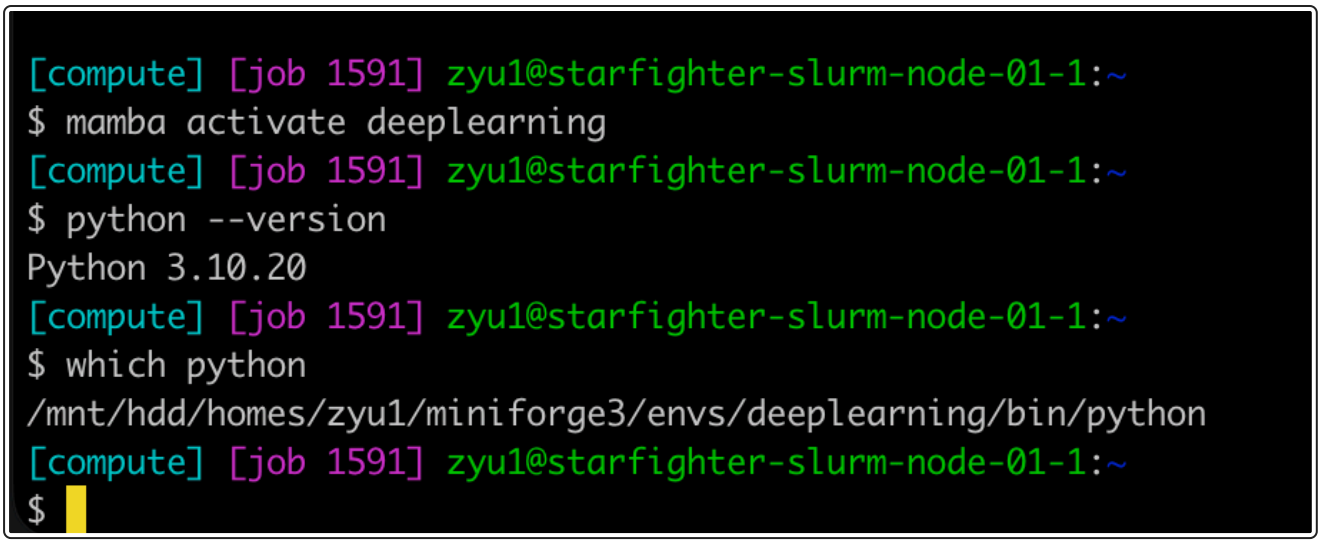
sent 80 bytes  received 26 bytes  5856 bytes/sec
total size is 0  speedup is 0.00
```

Ex2

2.a

```
[compute] [job 1591] zyu1@starfighter-slurm-node-01-1:~
$ mamba --version
2.6.0
```

2.b



pour vérifier la version exacte de Python: python --version
resultat: Python 3.10.20
le chemin du binaire utilisé: which python
resultat: /mnt/hdd/homes/zyu1/miniforge3/envs/deeplearning/bin/python

2.d

la sortie de ce script:
PyTorch version: 2.10.0
CUDA available: False
Attention, aucun GPU détecté !

Le version n'est pas correspondant.

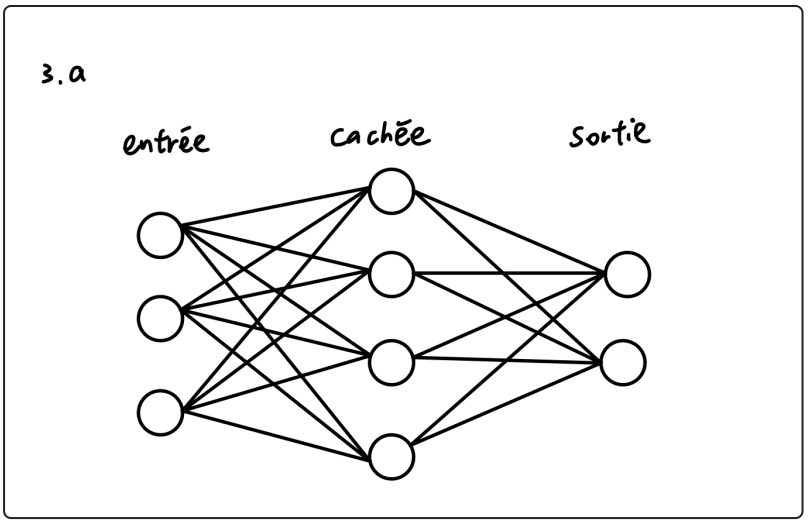
2.f



tensorboard --version

Ex 3

3.a



Sans biais:
couche 1 = 3 * 4 = 12
couche 2 = 4 * 2 = 8
le nombre total = couche 1 + couche 2 = 20

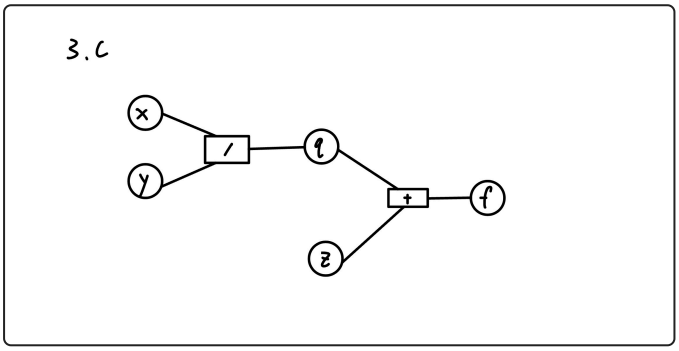
Avec biais:
biais 1 = 4
biais 2 = 2
le nombre total = couche 1 + biais 1 + couche 2 + biais 2 = 26

3.b

H = ReLU(X · W1^T + b1)
Y = H · W2^T + b2

Dimensions :
X : (N, 3)
W1 : (4, 3)
b1 : (1, 4) -> diffusé en (N, 4)
H : (N, 4)
W2 : (2, 4)
b2 : (1, 2) -> diffusé en (N, 2)
Y : (N, 2)

3.c



f = x / y + z = 0.5

∂f/∂x = ∂f/∂q * ∂q/∂x = 1/y * 1 = 0.25
∂f/∂y = ∂f/∂q * ∂q/∂y = -x/y^2 * 1 = -0.125
∂f/∂z = 1

3.d

x' = x - η * ∂f/∂x = 1.75
y' = y - η * ∂f/∂y = 4.125
z' = z - η * ∂f/∂z = -1
f' = f(x', y', z') = -0.576
Oui.

3.e

3.e.1
Parce que c'est plus pratique de calculer le gradient local.
3.e.2
C'est plus stable et peut exploiter le capacité de parallèle comparant sur un seul exemple à la fois.
C'est plus rapide est requit moins de ressources que sur l'ensemble total des données.

3.f

Tâche	Fonction finale (Sortie)	Fonction de perte (Loss)
Classification binaire	1. Sigmoid	A. Binary Cross-Entropy
Classification multi	2. Softmax	B. Cross-Entropy
Régression pure	3. Identité (aucune)	C. MSE (Mean Squared Error)

Ex4

4.a

batch_size: Nombre d'exemples participé à calculation chaque itération, pour faire tradeoff entre efficacité et ressources.
shuffle: Pour faire shuffle.
différence entre train et test: Pour training, il faut avoir samples différents, pour avoir mieux généralisation. Pour testing, fixer le séquences est plus facile pour reproducer le résultat.

4.b

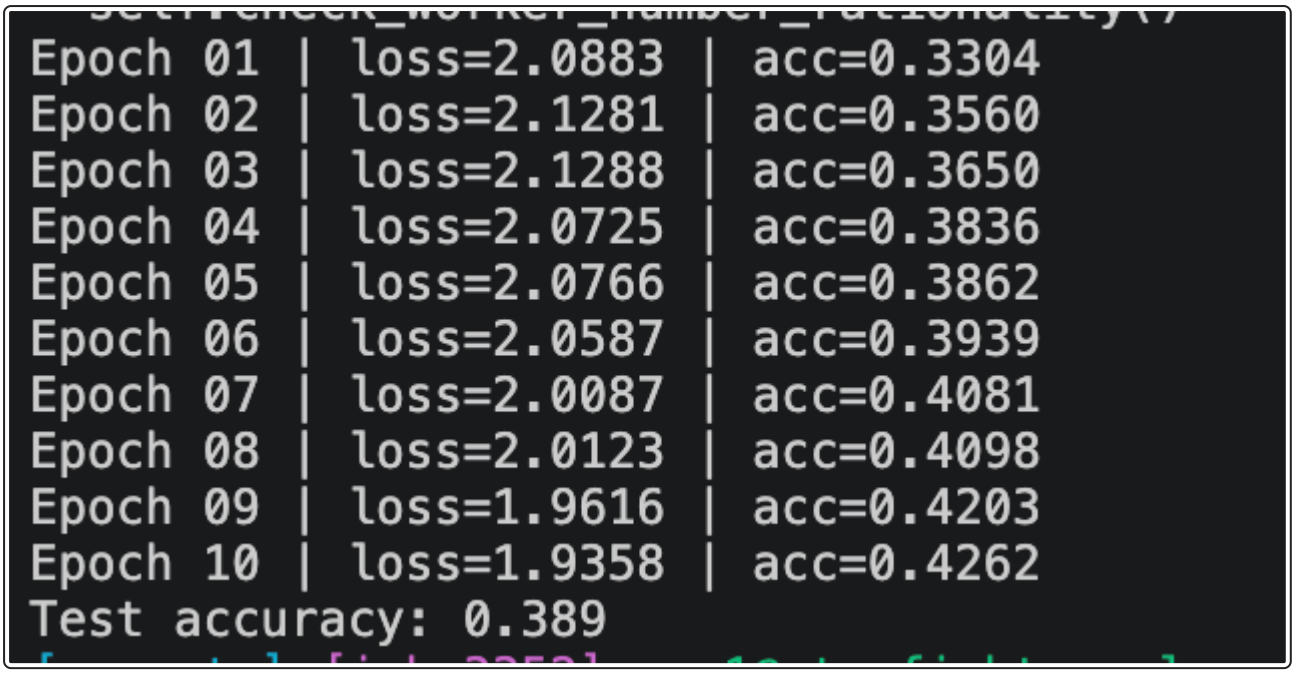
4.b.1 Pour correspondre le format d'input.
4.b.2 Il y a déjà un fonction d'activation dans nn.CrossEntropyLoss

4.c

optimizer.zero_grad(): reset the grad to 0
loss.backward(): calculate the new grad

4.d

4.d.1 On n'a pas besoin de mettre à jour le grad. Cela prend moins de ressources.
4.d.2 0.1



4.e

ok.

Ex5

5.a

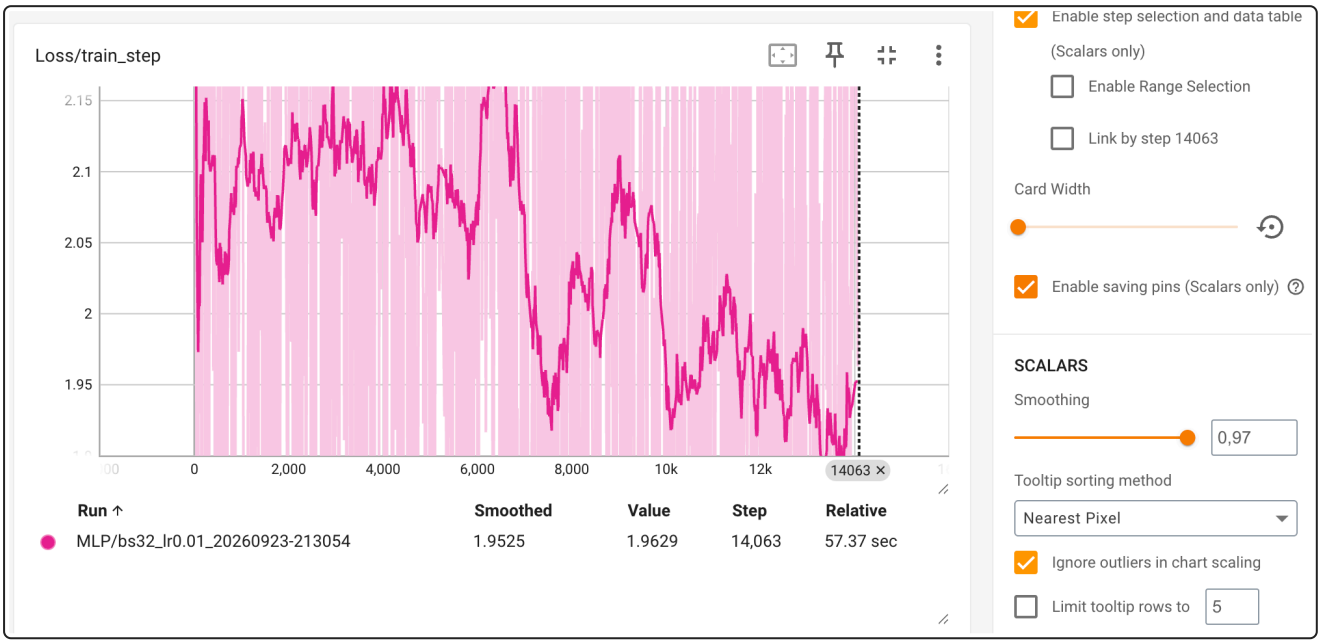
Pour savoir c'est quel expérience et on a faire quoi dans l'expérience.

5.d

```
$ python train_tb.py
Logdir: runs/MLP/bs32_lr0.01_20260923-213054
Using device: cuda
Epoch 01 | train_loss=2.0769 | val_loss=2.1728 | val_acc=0.327
Epoch 02 | train_loss=2.1167 | val_loss=2.1323 | val_acc=0.340
Epoch 03 | train_loss=2.1043 | val_loss=2.2177 | val_acc=0.363
Epoch 04 | train_loss=2.0809 | val_loss=2.2645 | val_acc=0.340
Epoch 05 | train_loss=2.0950 | val_loss=2.2803 | val_acc=0.358
Epoch 06 | train_loss=2.0598 | val_loss=2.1279 | val_acc=0.376
Epoch 07 | train_loss=2.0453 | val_loss=2.3534 | val_acc=0.365
Epoch 08 | train_loss=2.0041 | val_loss=2.2729 | val_acc=0.379
Epoch 09 | train_loss=1.9839 | val_loss=2.1524 | val_acc=0.391
Epoch 10 | train_loss=1.9284 | val_loss=2.3094 | val_acc=0.373
```

```
→  CSC8607 scp -r tsp-client:~/CSC8607/TP1/runs ./
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
events.out.tfevents.1790198655.starfighter-sl 100%   88     5.6KB/s   00:00
events.out.tfevents.1790198573.starfighter-sl 100%   88     5.5KB/s   00:00
events.out.tfevents.1790199054.starfighter-sl 100%   76KB 602.7KB/s   00:00
```

```
(.venv) →  CSC8607 tensorboard --logdir=runs
TensorFlow installation not found - running with reduced feature set.
Serving TensorBoard on localhost; to expose to the network, use a proxy or pass
--bind_all
TensorBoard 2.21.0 at http://localhost:6006/ (Press CTRL+C to quit)
```

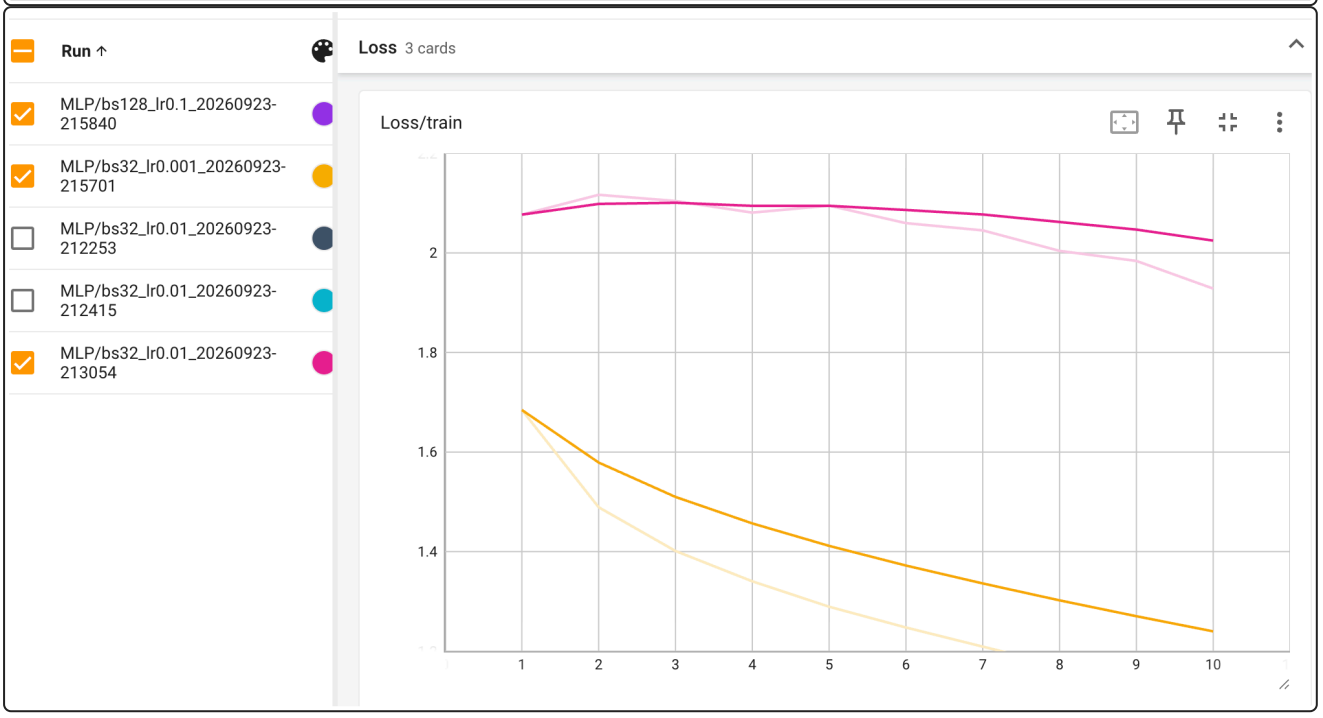


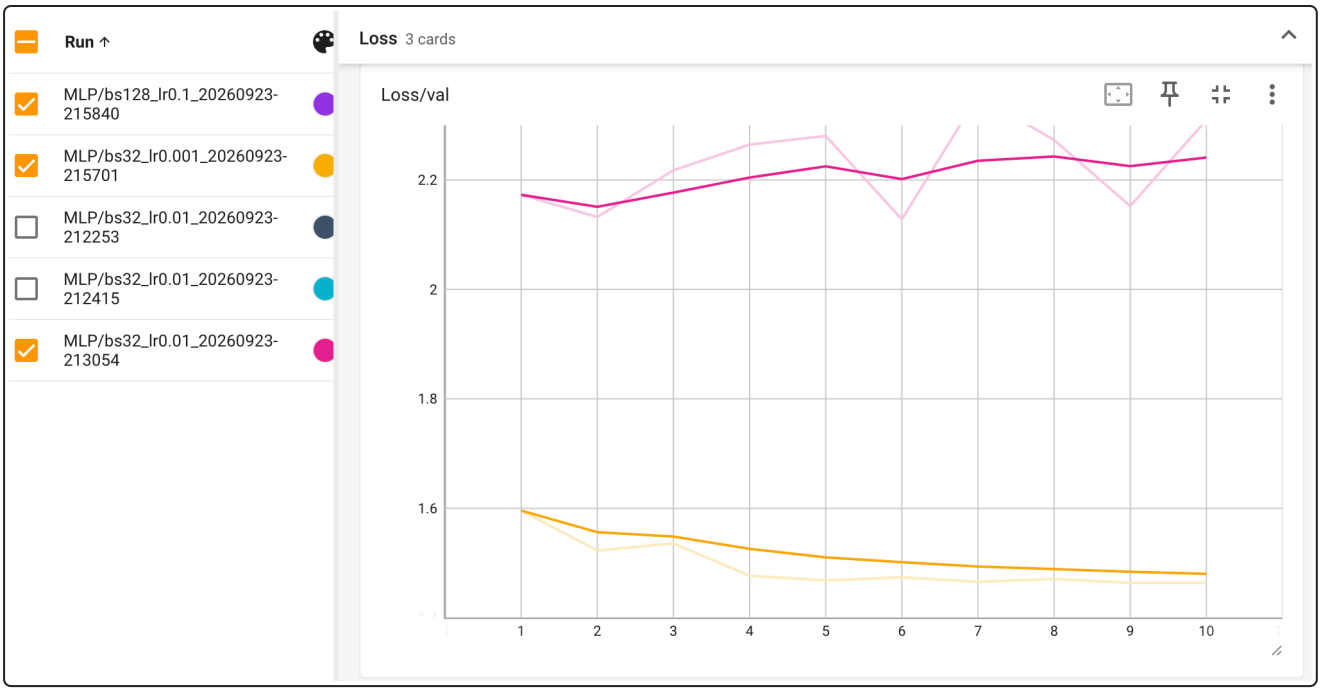
Vers 0.97

Loss/train_step est de batch de 32 exemples, qui est comparativement très petit donc a plus de bruit.

5.e

```
[compute] [job 3380] zyu1@starfighter-slurm-node-01-1:~/CSC8607/TP1 (main)
$ python train_tb.py
Logdir: runs/MLP/bs32_lr0.001_20260923-215701
Using device: cuda
Epoch 01 | train_loss=1.6846 | val_loss=1.5960 | val_acc=0.438
Epoch 02 | train_loss=1.4892 | val_loss=1.5229 | val_acc=0.473
Epoch 03 | train_loss=1.4014 | val_loss=1.5360 | val_acc=0.473
Epoch 04 | train_loss=1.3404 | val_loss=1.4770 | val_acc=0.487
Epoch 05 | train_loss=1.2894 | val_loss=1.4683 | val_acc=0.496
Epoch 06 | train_loss=1.2477 | val_loss=1.4740 | val_acc=0.495
Epoch 07 | train_loss=1.2093 | val_loss=1.4658 | val_acc=0.497
Epoch 08 | train_loss=1.1714 | val_loss=1.4712 | val_acc=0.499
Epoch 09 | train_loss=1.1391 | val_loss=1.4639 | val_acc=0.508
Epoch 10 | train_loss=1.1074 | val_loss=1.4639 | val_acc=0.515
[compute] [job 3380] zyu1@starfighter-slurm-node-01-1:~/CSC8607/TP1 (main)
$ python train_tb.py
Logdir: runs/MLP/bs128_lr0.1_20260923-215840
Using device: cuda
Epoch 01 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 02 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 03 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 04 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 05 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 06 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 07 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 08 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 09 | train_loss=nan | val_loss=nan | val_acc=0.096
Epoch 10 | train_loss=nan | val_loss=nan | val_acc=0.096
[compute] [job 3380] zyu1@starfighter-slurm-node-01-1:~/CSC8607/TP1 (main)
$
```





5.e.1 Run 2

5.e.2 Training loss continue de diminuer, et validation loss diminue puis remonte.